

Task 11 — Offer Generation & E-Sign Design

Data Analyst • Week 4 • Phase 2

1. Objective

Build an offer funnel that traces interviewed candidates through offer generation, offer sending, e-signature and hiring. Every metric is tied to row-level application/offer records.

2. Dataset & Source

The base application funnel contains 1,000 application records, including 270 interviewed candidates and 85 offers. The Task 11 CSV extends the offer records with synthetic offer and e-sign fields. The extension is clearly labeled as synthetic because the original application dataset does not contain e-sign events.

3. Funnel KPIs

Metric	Result
Applications	1,000
Interviewed	270
Offers Generated	85
Interview → Offer	31.5%
E-Sign Sent	85
Signed	59
Pending	12
Declined	14
E-Sign Completion	69.4%
Offer → Hire	69.4%

4. Offer Funnel Design

Interviewed → Offer Generated → Offer Sent → E-Sign Sent → Signed/Declined/Pending → Hired. The dashboard should show both counts and conversion rates, with candidate/application IDs available for drill-down.

5. E-Sign Decision Metrics

E-sign completion rate = $\text{signed offers} \div \text{offers sent} \times 100$. Pending offers should be monitored against an agreed SLA. Decline reasons should be grouped to identify compensation, role-fit or process issues.

6. Data Quality & Trust Checks

Checks should cover unique application IDs, unique offer IDs, non-null offer IDs for offers, valid e-sign status values, chronological offer/e-sign timestamps, and logical funnel ordering. The synthetic e-sign extension should remain labeled until real e-sign events are integrated.

7. Business Insights

- 85 offers were generated from 270 interviewed candidates (31.5% interview-to-offer).
- 59 offers are signed, giving a 69.4% e-sign completion rate.
- 12 offers remain pending and should be monitored using an SLA.
- 14 offers are marked declined; decline reasons provide a direct improvement signal.
- 59 offer recipients are marked hired, giving a 69.4% offer-to-hire rate.

8. Recommended Dashboard

KPI cards: interviewed, offers, e-sign sent, signed, pending, declined and hired. Funnel chart: interview → offer → e-sign → hire. Additional views: source comparison, decline reasons, pending-offer aging, role/company filters, and candidate-level traceability.

9. Definition of Done

Offer funnel metrics are defined, the e-sign path is measurable, the data is reproducible from CSV records, sanity checks are included, and every displayed number can be explained to a decision-maker.

Important source note: The application funnel is based on the earlier synthetic application dataset. Offer/e-sign fields added for Task 11 are synthetic extensions for demonstrating the required design and analytics pipeline.